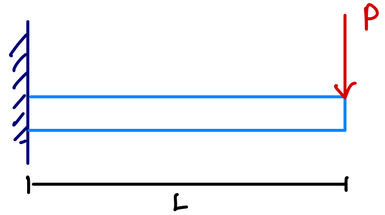
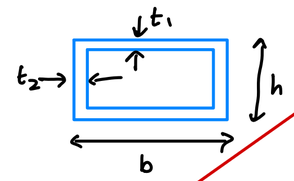
OPTIMIZATION PROBLEM FORMULATION





$$A_c = h \cdot b - \left[(h-2t_1)(b-2t_2)\right]$$

$$\delta_{MAX}(\vec{x}) = \frac{PL^3}{3EI_{zz}} \qquad I_{zz} = \frac{1}{12}bh^3 - \frac{1}{12}(b-2t_2)(h-2t_1)^3$$

$$m = L \cdot A_c \cdot \rho \qquad \sigma_{max} = \frac{My}{I_{zz}} = \frac{(P \cdot L)(y)}{I_{zz}}$$

$$f(\vec{x}) = m(\vec{x}) = \rho \cdot L \cdot A_c(\vec{x}) = \rho \cdot L \cdot \left[(h \cdot b) - \left[(h-2t_1)(b-2t_2)\right]\right]$$

$$\vec{x} = \begin{Bmatrix} b \\ h \\ t_1 \\ t_2 \end{Bmatrix} \qquad \begin{matrix} b = x_1 \\ h = x_2 \\ t_1 = x_3 \\ t_2 = x_4 \end{matrix}$$

CONSTRAINTS

$0.1 \le h \le 1$ ← LOWER + UPPER LIMITS ON h

$0.1 \le b \le 1$ ← LOWER + UPPER LIMITS ON b

$0.01 \le t_1 \le 0.25$ ← LOWER + UPPER LIMITS ON t_1

$0.01 \le t_2 \le 0.25$ ← LOWER + UPPER LIMITS ON t_2

$\delta_L \le 0.5$ ← LOWER + UPPER BOUNDS ON δ_{MAX}

$\sigma_{max} \le 1.5\sigma_y$ ← UPPER BOUND ON STRESS

OPTIMIZATION PROBLEM FORMULATION

$$\vec{x} = \begin{Bmatrix} b \\ h \\ t_1 \\ t_2 \end{Bmatrix}$$

$$\min_{\vec{x} \in \mathbb{R}^4} f(\vec{x}) = m(\vec{x}) = \rho L A_c(\vec{x}) = \rho L \left[(x_2 - x_1) - \left[(x_2 - 2x_3)(x_1 - 2x_4)\right]\right]$$

S.T.

$0.1 \le x_2 \le 1$

$0.1 \le x_1 \le 1$

$0.01 \le x_3 \le 0.25$

$0.01 \le x_4 \le 0.25$

$\delta_L \le 0.5$

$\sigma_{max} \le 1.5\sigma_y$

$$\delta_L(\vec{x}) = \frac{PL^3}{3EI_{zz}(\vec{x})} \qquad I_{zz}(\vec{x}) = \frac{1}{12}x_1 x_2^3 - \frac{1}{12}(x_1 - 2x_4)(x_2 - 2x_3)^3$$

$$\sigma_{max}(\vec{x}) = \frac{(PL)(\frac{x_2}{2})}{I_{zz}(\vec{x})}$$

#10 INEQUALITY CONSTRAINTS

#WELL-DEFINED PROBLEM

POSSIBLE SOLUTION TECHNIQUES

* START W/ BARRIER METHOD
* LAGRANGE MULTIPLIERS
* SQP
  ↳ SEQUENTIAL QUADRATIC PROGRAMMING
* FMINCON - PROVIDE REFERENCE SOLUTION TO COMPARE WITH